DATA INTEGRITY POLICY

POL-003 | Issued under the Information Management Framework

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Revision History

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0.1	[DATE]	[Author]	Initial draft
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1. Purpose

This Policy establishes mandatory requirements for maintaining the integrity of data across all activities at [Company Name] that are subject to Good Practice (GxP) regulations, and more broadly across all regulated and critical business data. It applies the internationally recognised ALCOA+ framework as the governing standard for data integrity at [Company Name].

Data integrity failures are among the most frequently cited causes of regulatory action against life-sciences companies globally — including FDA warning letters, import alerts, consent decrees, and marketing authorisation withdrawals. This Policy exists to prevent those failures by embedding data integrity as a foundational operating principle rather than a compliance afterthought.

> **GxP:** Data integrity is not solely a GxP concern. This Policy applies to any data that [Company Name] relies upon to make decisions, submit to regulators, present to auditors, or use in legal proceedings. GxP data carries the highest obligation, but the principles apply universally.

2. Scope

This Policy applies to:

- All GxP data: any data created, modified, maintained, archived, retrieved, or transmitted in support of regulated activities (GMP, GLP, GCP, GDP, GPvP)
- All electronic records and electronic signatures subject to 21 CFR Part 11 or EU Annex 11
- All SOX-relevant financial records and IT general control evidence
- All data classified as Tier 3 (Sensitive) or above under POL-001, wherever data integrity is material to the use of that data
- All personnel who create, review, approve, modify, archive, or retrieve regulated data
- All systems — internal and third-party — that hold, process, or generate GxP or other regulated records

Where this Policy references GxP specifically, it applies to all Good Practice contexts (GMP, GLP, GCP, GDP, GPvP) unless a specific sub-regulation is cited.

3. The ALCOA+ Standard

ALCOA+ is the internationally recognised framework for data integrity in the life-sciences sector, endorsed by the FDA, EMA, MHRA, WHO, and ICH. It defines the attributes that all GxP

records must possess. At [Company Name], ALCOA+ is the mandatory standard for all data within the scope of this Policy.

Attribute	Definition	Practical Obligation at [Company Name]
A — Attributable	It must be possible to determine who created, modified, or reviewed the record, and when.	All records must be linked to the individual who created or modified them — through login credentials, electronic signatures, or physical signature. Generic or shared logins are prohibited for GxP systems. System audit trails must capture user identity.
L — Legible	The record must be readable and permanent throughout its required retention period.	Data must be recorded in a clear, durable format. Electronic records must be viewable in human-readable form throughout the retention period. Degraded, overwritten, or illegible records are non-compliant. Abbreviations must be defined in a controlled glossary.
C — Contemporaneous	The record must be created at the time the activity occurs — not reconstructed after the fact.	Data must be recorded at the time of observation or action, not retrospectively. Pre-dating, back-dating, or reconstructing records is prohibited. Where records are created in electronic systems, timestamps must be system-generated, not manually entered.
O — Original	The first capture of data must be preserved. Original records must not be replaced by transcribed copies unless the original is also retained.	Raw data — the first record of an observation, result, or action — must be preserved. Transcription into a secondary record is permitted only when the original is retained and the transcription is verified. Electronic originals must not be routinely converted to paper.
A — Accurate	The record must correctly reflect what was observed, measured, or done. It must be free from errors and bias.	Records must reflect actual observations and events. Rounding, estimating, or correcting values without documented justification is prohibited. Instruments and systems producing data must be calibrated and qualified. Corrections must preserve the original and be explained.
+ Complete	All data — including repeats, reanalyses, and out-of-specification results — must be retained. Nothing may be selectively omitted.	All generated data must be captured and retained, including failed runs, OOS results, and abandoned experiments. Selective reporting or cherry-picking of results is a serious integrity violation. Audit trails must capture all data entry and modification events.
+ Consistent	Procedures and practices must be applied uniformly. Records must be internally coherent.	Processes must be performed consistently per approved procedures. Discrepancies between raw data, reports, and submissions must be reconciled and documented. System clocks must be synchronised to a reliable time source.
+ Enduring	Records must be maintained in a durable, unalterable form for the required retention period.	Records must be stored in a format that cannot be altered without detection. Electronic records must be stored in validated systems with controlled access and immutable audit trails. Paper records must be stored in conditions that prevent deterioration.

Attribute	Definition	Practical Obligation at [Company Name]
+ Available	Records must be retrievable and accessible to authorised personnel — including regulators — throughout the retention period.	Records must be stored in a manner that allows timely retrieval on request. Regulatory inspectors must be able to access records within the timeframe stipulated in the inspection notice. Archive systems must be tested for retrieval capability.

4. Mandatory Requirements

4.1 Electronic Records and Audit Trails

All GxP electronic records must be maintained in validated systems (per POL-006 — Computer System Validation and Qualification Policy). The following requirements apply to all electronic GxP records:

- Audit trails must be enabled and configured to capture: user identity, date and time of action (system-generated), original value, new value, and reason for change
- Audit trail data must be stored in a format that cannot be modified, overwritten, or deleted by system users — including system administrators
- Audit trails must be reviewed as part of routine quality oversight — the frequency and scope of review must be defined in the relevant SOP
- System clocks must be synchronised to a validated, tamper-resistant time source (e.g., NTP server) and changes to system time must be logged
- Access to audit trail data must be restricted — only authorised quality or compliance personnel may review audit trails; no operational user may modify them

IMPORTANT: Disabling, bypassing, or modifying audit trail configuration in a GxP system is a critical data integrity violation and will be treated as a serious breach subject to disciplinary action and regulatory notification where required.

4.2 Electronic Signatures

Where [Company Name] uses electronic signatures in place of handwritten signatures on GxP records, the following requirements apply under 21 CFR Part 11 and EU Annex 11:

- Electronic signatures must be unique to one individual and must not be reused or reassigned
- Each electronic signature must link permanently to the record it authenticates
- Electronic signatures must include: the printed name of the signer, the date and time of signing, and the meaning of the signature (e.g., reviewed, approved, performed)
- The use of electronic signatures must be authorised in writing to the FDA (where applicable) prior to implementation (21 CFR 11.100)
- Systems using electronic signatures must implement controls to prevent signature falsification

4.3 Access Controls and Identity

Identity controls are the primary mechanism for ensuring the Attributable requirement of ALCOA+. The following must be enforced in all GxP systems:

- Each user must have a unique, individual login — generic, shared, or role-based accounts that are used by multiple individuals are prohibited

- Password controls must meet the standards defined in POL-008 (Access Control Policy)
- Users must not share login credentials under any circumstances
- Automatic screen lock and session timeout must be configured on all GxP system interfaces
- Access must be granted on a need-to-know basis and reviewed per the frequency defined in POL-008 for the system's criticality tier

4.4 Data Entry and Recording

The following practices are mandatory for all personnel entering data into GxP records:

- Data must be recorded at the time of the activity — not from memory or notes made later
- All entries must be made directly into the controlled record — temporary recording on scraps of paper or unofficial notebooks, followed by later transcription, is prohibited unless the original is retained and the transcription verified
- Errors in paper records must be corrected by: crossing out the original entry with a single line (not obscuring it), writing the correct value, dating and initialling the correction, and noting the reason
- Errors in electronic records must be corrected using the system's controlled amendment function — original values must be preserved in the audit trail
- Blank spaces in paper records must be marked as not applicable (N/A) or left intentionally blank with a documented reason — blank fields must never be left ambiguous

4.5 Data Transfer and Migration

When data is transferred between systems — whether as part of integration, migration, or archival — the following must be ensured:

- A validated data transfer process must be in place, with documented verification that transferred data is complete, accurate, and has not been altered in transit
- Original records must not be destroyed until the receiving system has been validated and the data verified
- Data migration must be covered by the system validation programme (POL-006) and must include reconciliation of record counts and spot-check verification of content
- Audit trail data must be migrated alongside the records it relates to

4.6 Hybrid and Paper-Electronic Systems

Where [Company Name] operates hybrid systems — combining paper and electronic records for the same process — the following additional requirements apply:

- The master record type (paper or electronic) must be defined in the relevant SOP and applied consistently
- Where paper is the original, any electronic transcription is a copy — the paper original must be retained

- Where electronic is the original, printed copies are not controlled records — they must be marked as uncontrolled copies
- Hybrid processes must be assessed as part of the Data Integrity Risk Assessment (Section 5) — they typically present elevated data integrity risk

5. Data Integrity Risk Assessment

All GxP systems and processes must be subject to a Data Integrity Risk Assessment (DIRA). The DIRA identifies the points in a process where data integrity vulnerabilities exist, and defines controls to mitigate them. DIRAs are a regulatory expectation and will be requested during inspection.

DIRA Component	Description	Frequency
Process mapping	Document the end-to-end data flow for the process, identifying all points where data is created, transferred, reviewed, approved, or archived.	At process design or significant change
Vulnerability identification	For each data point, assess the risk that data could be: lost, altered without detection, entered by an unauthorised person, entered retrospectively, or selectively omitted.	Per process map
Control assessment	Evaluate existing controls against each vulnerability. Identify gaps.	Per process map
Risk rating	Rate each vulnerability (residual risk post-controls) as Critical, High, Medium, or Low based on likelihood and impact.	Per process map
Remediation	Define additional controls for Critical and High residual risks with target completion dates and owners.	Per risk rating
DIRA review	Review and update the DIRA when the process, system, or regulatory landscape changes materially, or at least every 3 years.	Every 3 years or on change

DIRAs must be documented, QA-approved, and retained for the life of the process plus the applicable GxP record retention period. They are subject to inspection by regulatory authorities.

6. Governance and Oversight

6.1 Data Integrity Officer

[Company Name] designates a Data Integrity Officer (DIO), typically the Head of Quality Assurance or a senior QA designee. The DIO is responsible for:

- Overall programme management of data integrity across the organisation
- Overseeing the DIRA programme and ensuring completion across all GxP systems
- Reviewing audit trail reports and escalating anomalies

- Maintaining this Policy and its subordinate SOPs
- Acting as the primary point of contact for regulatory inspectors on data integrity matters
- Reporting on data integrity posture to senior leadership quarterly

6.2 Roles and Responsibilities

Role	Data Integrity Responsibility
All Personnel	Record data accurately, contemporaneously, and completely. Never alter records without authorisation. Report suspected data integrity issues immediately.
Line Managers	Ensure their teams understand and comply with this Policy. Do not exert pressure that could compromise data integrity (e.g., pressure to meet targets at the expense of accurate recording).
Data Integrity Officer	Programme management, DIRA oversight, audit trail review, regulatory liaison, policy maintenance, incident investigation lead.
Quality Assurance	Independently review data integrity compliance across GxP activities. Review audit trails. Conduct data integrity audits. Approve DIRAs.
Information Security	Ensure system controls support data integrity requirements: access management, audit trail configuration, backup integrity, system clock controls.
IT / System Owners	Implement and maintain technical controls mandated by this Policy. Ensure audit trail functionality is enabled and not suppressible. Maintain system validation status.

7. Training

All personnel who create, review, approve, or handle GxP records must complete data integrity training before working with regulated data, and at least annually thereafter. Training must cover:

- The ALCOA+ standard and what it means in practice for their role
- Prohibited practices — back-dating, pre-signing, shared logins, selective data retention
- How to make corrections correctly — paper and electronic
- How to report suspected data integrity issues
- The consequences of data integrity failures — regulatory, professional, and personal

Training completion and assessment must be documented and retained per POL-002 (Data Retention and Disposal Policy). Training records are subject to regulatory inspection.

8. Data Integrity Incidents and Investigations

Any suspected or confirmed data integrity failure — including retrospective data entry, audit trail manipulation, falsification of records, or selective data deletion — must be reported immediately to the Data Integrity Officer and Quality Assurance.

Severity	Definition	Response
Critical	Evidence or strong suspicion of deliberate data falsification, fraud, or audit trail manipulation.	Immediate escalation to CISO, Legal, and senior leadership. Incident investigation initiated within 24 hours. Regulatory notification assessed. Potential law enforcement referral.
High	Systemic data integrity failures across a process or system, or a single serious failure affecting a regulatory submission or patient safety.	Formal CAPA initiated within 5 business days. QA-led investigation. Regulatory authority notification assessed. Affected records quarantined pending investigation.
Medium	Isolated non-compliance with ALCOA+ requirements, without evidence of intent, and not affecting regulatory submissions.	CAPA initiated within 30 days. Root cause analysis. Retraining of affected personnel. Record review and remediation.
Low	Minor procedural non-compliance identified during routine audit or self-inspection, with no impact on data reliability.	Documented observation. Corrective action within 90 days. Tracked to closure by QA.

All data integrity incidents, investigations, and CAPAs must be documented and retained per POL-002. The Data Integrity Officer provides a quarterly summary of incidents and trends to the ISSG.

9. Related Documents

Document ID	Title	Relationship
IMF-001	Information Management Framework	Parent framework
POL-001	Data Classification Policy	GxP flag assigned under POL-001 governs application of this Policy
POL-002	Data Retention and Disposal Policy	Retention periods for GxP records; audit trail records retained alongside primary records

Document ID	Title	Relationship
POL-005	System Classification Policy	GxP systems classified per POL-005; this Policy sets additional technical requirements for those systems
POL-006	Computer System Validation and Qualification Policy	Validation requirement for GxP systems; ALCOA+ controls validated as part of CSV/CSA
POL-007	Change Management Policy	Uncontrolled change to GxP systems is the primary cause of data integrity failures; all changes to GxP systems must go through POL-007 and include a validation impact assessment
POL-008	Access Control Policy	Access controls support Attributable requirement; individual logins and access review cadence
POL-010	Information Security Risk Policy	DIRA is the GxP domain-specific instance of the IS risk framework; DIRA Critical and High findings must be reflected in the organisation-wide Risk Register maintained under POL-010; DIRA Register is a sub-register of the IS Risk Register
POL-011	Business Continuity Policy	GxP system BC/DR plans must address validated state restoration, not just operational restoration; audit trail continuity during DR events is a data integrity obligation under this Policy; manual backup procedures required for all GxP systems
POL-015	IT Service Management and Operational Incident Policy	Unplanned operational outages of GxP systems create audit trail gaps that are ALCOA+ Completeness failures under this Policy; the incident period must be documented in the audit trail; validated state must be confirmed before GxP restart
EXC-001	Exception Management Policy and Register	Exceptions to GxP data integrity requirements (ALCOA+ attributable and complete) are Category 2 under EXC-001 with mandatory Compliance/QA involvement; shared accounts on GxP systems are non-approvable per EXC-001 Section 5
POL-012	Incident Response Policy	Data integrity incidents — audit trail manipulation, unauthorised record modification, uncontrolled configuration change — are security incidents governed by POL-012; Compliance / QA must be involved in all incident response for GxP systems; no GxP system may return to regulated use after an incident without validated state confirmation
POL-014	Audit Trail and Monitoring Policy	Audit trail configuration and review requirements referenced by this Policy

Document ID	Title	Relationship
SOP-003	Data Integrity Process	Operationalises this Policy (planned)
RACI-001	Roles, Responsibilities and Authorities	Defines accountability for data integrity activities

10. Definitions

Term	Definition
ALCOA+	The data integrity standard mandated for GxP records: Attributable, Legible, Contemporaneous, Original, Accurate, Complete, Consistent, Enduring, Available.
Audit Trail	A secure, time-stamped record of system and user activities that allows reconstruction of data creation, modification, and deletion events.
CAPA	Corrective Action and Preventive Action — the documented process for addressing the root cause of a non-compliance and preventing recurrence.
DIRA	Data Integrity Risk Assessment — a structured evaluation of data integrity vulnerabilities in a process or system, with controls and residual risk ratings.
Electronic Record	Any combination of text, graphics, data, audio, pictorial, or other information representation in digital form created, modified, maintained, archived, retrieved, or distributed by a computer system.
Electronic Signature	A computer data compilation of any symbol or series of symbols executed, adopted, or authorised by an individual to be the legally binding equivalent of the individual's handwritten signature.
GxP	The collective term for Good Practice quality guidelines and regulations applicable to life-sciences: GMP (Good Manufacturing Practice), GLP (Good Laboratory Practice), GCP (Good Clinical Practice), GDP (Good Distribution Practice), GPvP (Good Pharmacovigilance Practice).
Raw Data	The first original record of a primary observation, measurement, or action — whether recorded on paper or electronically.

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